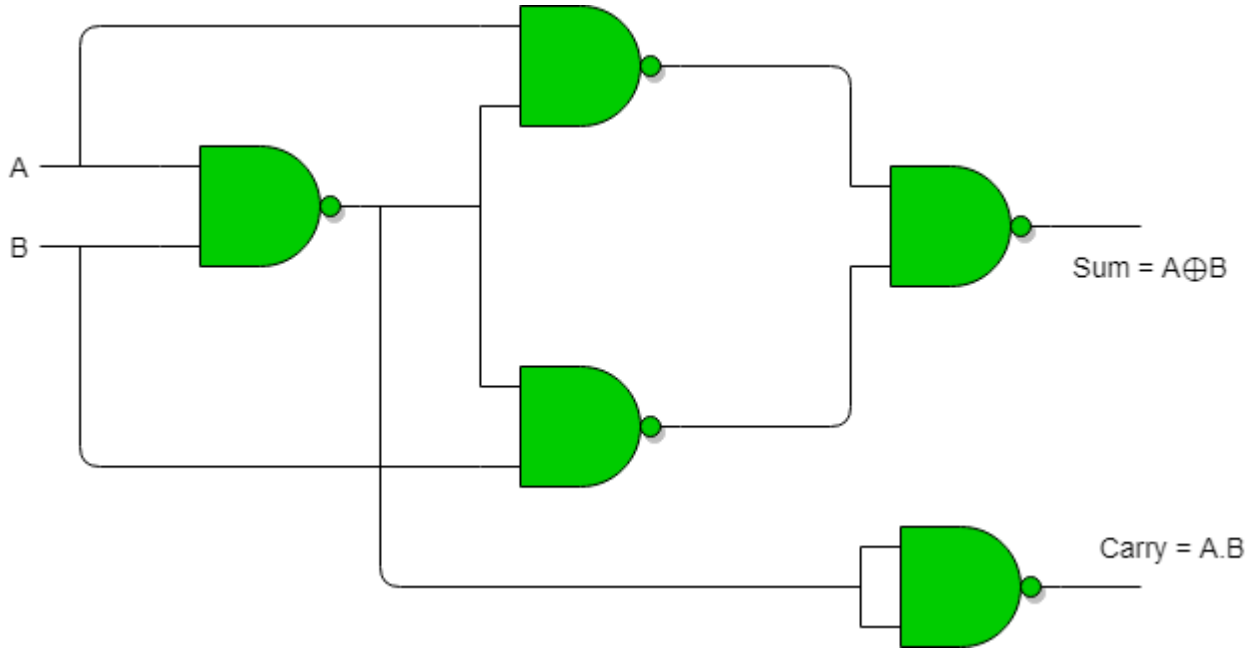
EXPERIMENT-8

HALF ADDER AND HALF SUBSTRACTOR-UNIVERSAL GATES

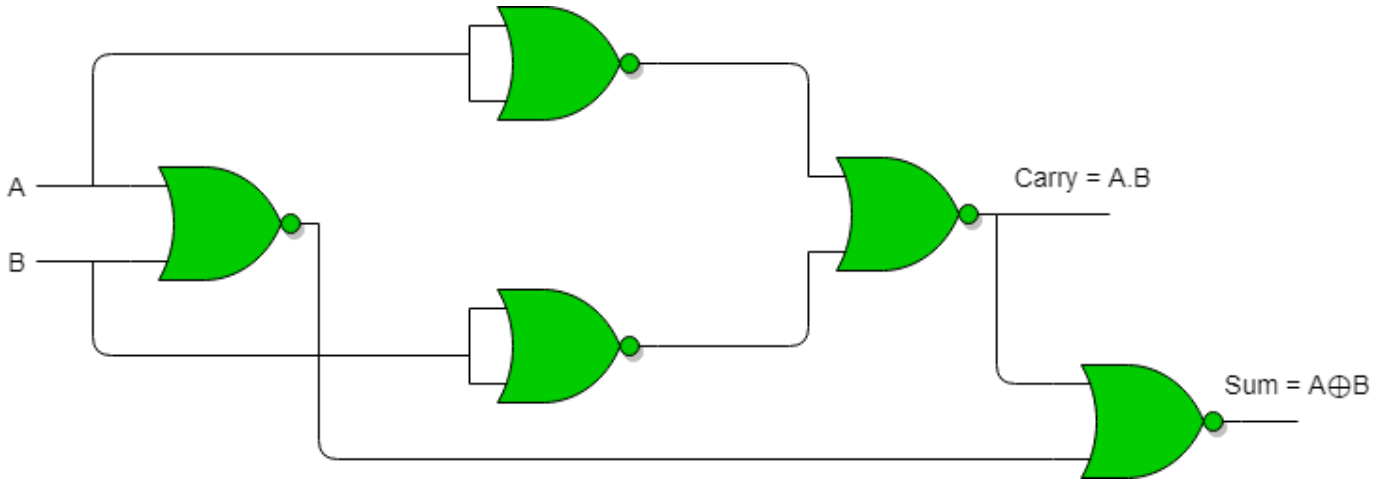
AIM: Implement half adder and half substractor using NAND and NOR gates

COMPONENTS REQUIRED: IC 7400,7402

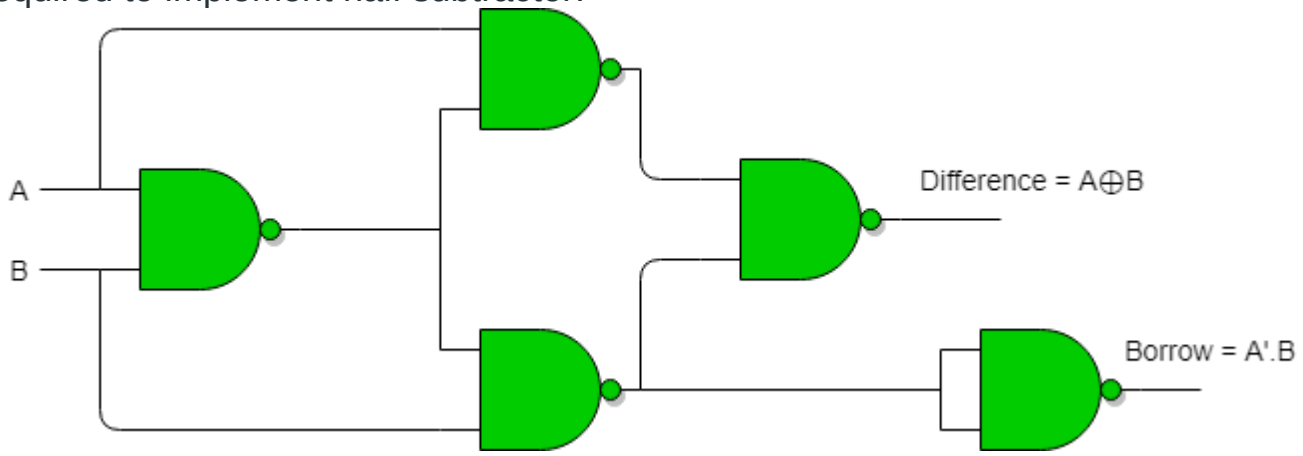
THEORY: Universality: NAND and NOR gates are considered universal gates because they can be used to implement any logical function, including binary arithmetic functions such as addition and subtraction.
Implementation of Half Adder using NAND gates : Total 5 NAND gates are required to implement half adder.



Implementation of Half Adder using NOR gates : Total 5 NOR gates are required to implement half adder.



Implementation of Half Subtractor using NAND gates : Total 5 NAND gates are required to implement half subtractor.



Implementation of Half Subtractor using NOR gates : Total 5 NOR gates are required to implement half subtractor.

